COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010- Data Structures and Algorithms	**SECTION:**	CPE21S4
DATE PERFORMED:	10/02/2026	**DATE SUBMITTED:**	10/02/2026
NAME:	Irish Crystle C. Prieto Jullian Edrain Sevilla	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?

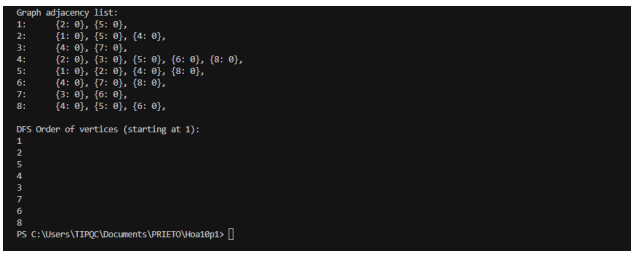
2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

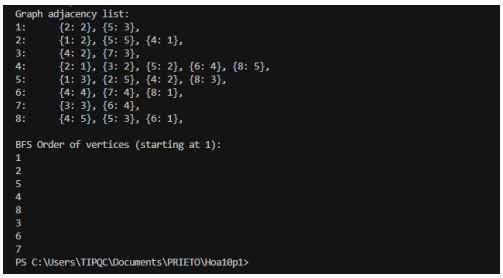
ILO A:



ILO B:



B.2



Supplementary Activity
ILO C:

1. A person wants to visit different locations indicated on a map. He starts from one location (vertex) and wants to visit every vertex until it finishes from one vertex, backtracks, and then explore other vertex from the same vertex. Discuss which algorithm would be most helpful to accomplish this task.

- Depth-First Search, or DFS, is the best algorithm for this task. It starts at one place and keeps going deeper until it cannot go further. Then it goes back step by step — this is called backtracking. After backtracking, it explores other paths from the same starting point. This matches exactly what the person does on the map.

2. Identify the equivalent of DFS in traversal strategies for trees. To efficiently answer this question, provide a graphical comparison, examine pseudocode and code implementation.

- For trees, the equivalent of DFS is called Pre-order Traversal. Both go deep into one branch before moving to the next.DFS on graphs marks nodes so they are not visited again. Trees have no loops, so we just visit each child one by one. We visit the node first, then its children, then backtrack. They work the same way — going down first before going across.

3. In the performed code, what data structure is used to implement the Breadth First Search?

- Breadth-First Search uses a Queue as its main data structure. A queue works like a line — first in, first out. We add new nodes to the end and take nodes from the front. This makes BFS visit all nearby nodes before moving further away. It explores the graph level by level.

4. How many times can a node be visited in the BFS?

- A node is visited only one time in BFS. Once we visit a node, we mark it so we do not visit it again. This keeps the search fast and stops repeating the same place. Every node gets processed exactly once.

Conclusion:
I learned that graphs show connections between points using DFS with a stack that goes deep first and backtracks, versus BFS with a queue that visits level by level. The code worked clearly once fixed. The questions helped me tell them apart and see how DFS matches tree pre-order traversal. I did okay understanding the main ideas, but I need to practice remembering their differences.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Grammarly**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **GRammar corrections**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **Didnot replace original thinking**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or

completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.

ILO A

```
Graph adjacency list
(start_vertex, end_vertex, weight):
(0, 2, 4) (0, 1, 2)
(1, 4, 3)
(2, 3, 2)
(3, 1, 4)
(4, 3, 3)

PS C:\Users\TIPQC\Documents\PRIETO\Hoa10p1>

(1, 4, 3)
(2, 3, 2)
(3, 1, 4)
(4, 3, 3)

PS C:\Users\TIPQC\Documents\PRIETO\Hoa10p1>
(1, 4, 3)
(2, 3, 2)
(3, 1, 4)
(4, 3, 3)
(1, 4, 3)
(2, 3, 2)
(3, 1, 4)
(1, 4, 3)
(1, 4, 3)
(2, 3, 2)
(3, 1, 4)
(4, 3, 3)

PS C:\Users\TIPQC\Documents\PRIETO\Hoa10p1>
```

ILO B

```
Graph adjacency list:
1:      {2: 0}, {5: 0},
2:      {1: 0}, {5: 0}, {4: 0},
3:      {4: 0}, {7: 0},
4:      {2: 0}, {3: 0}, {5: 0}, {6: 0}, {8: 0},
5:      {1: 0}, {2: 0}, {4: 0}, {8: 0},
6:      {4: 0}, {7: 0}, {8: 0},
7:      {3: 0}, {6: 0},
8:      {4: 0}, {5: 0}, {6: 0},

DFS Order of vertices (starting at 1):
1
2
5
4
3
7
6
8
PS C:\Users\TIPQC\Documents\PRIETO\Hoa10p1>
```

B.2

```
Graph adjacency list:
1:      {2: 2}, {5: 3},
2:      {1: 2}, {5: 5}, {4: 1},
3:      {4: 2}, {7: 3},
4:      {2: 1}, {3: 2}, {5: 2}, {6: 4}, {8: 5},
5:      {1: 3}, {2: 5}, {4: 2}, {8: 3},
6:      {4: 4}, {7: 4}, {8: 1},
7:      {3: 3}, {6: 4},
8:      {4: 5}, {5: 3}, {6: 1},

BFS Order of vertices (starting at 1):
1
2
5
4
8
3
6
7
PS C:\Users\TIPQC\Documents\PRIETO\Hoa10p1>
```

Grammarly:

Untitled document — Goals — **95** Overall score

Depth-First Search, or DFS, is the best algorithm for this task.
It starts at one place and keeps going deeper until it cannot go further.
Then it goes back step by step — this is called backtracking.
After backtracking, it explores other paths from the same starting point.
This matches exactly what the person does on the map.

Untitled document — Goals — **91** Overall score

For trees, the equivalent of DFS is called Pre-order Traversal. Both go deep into one branch before moving to the next.DFS on graphs marks nodes so they are not visited again. Trees have no loops, so we just visit each child one by one. We visit the node first, then its children, then backtrack. They work the same way — going down first before going across.

Untitled document

Goals 100 Overall score

Breadth-First Search uses a Queue as its main data structure. A queue works like a line — first in, first out. We add new nodes to the end and take nodes from the front. This makes BFS visit all nearby nodes before moving further away. It explores the graph level by level.

Untitled document

Goals 100 Overall score

A node is visited only once in BFS. Once we visit a node, we mark it so we do not visit it again. This keeps the search fast and stops repeating the same place. Every node gets processed exactly once